PROJECT PHASE-I REPORT

APPLICATION OF COIR GEOTEXTILE FOR SLOPE STABILIZATION IN EMBANKMENT

Submitted in partial fulfilment of the requirements
for the award of the degree of

BACHELOR OF TECHNOLOGY

in

CIVIL ENGINEERING

of the

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

by

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October 2026

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Thalakkottukara, Thrissur – 680 501

Department of Civil Engineering

Certificate

This is to certify that the project Phase-I report titled **APPLICATION OF COIR GEOTEXTILE FOR SLOPE STABILIZATION IN EMBANKMENT** is a bonafide record of the work carried out by **Anitha N. K. (VAS23CE011), Anupa V. J. (LVAS23CE048), K Ashika Musthafa (VAS23CE023) and M Sidharth Anand (VAS23CE028)** of Vidya Academy of Science & Technology, Thrissur, in partial fulfilment of the requirements for the award of the **Degree of Bachelor of Technology** in **Civil Engineering** of **APJ Abdul Kalam Technological University**, during the academic year 2026-2027. The report has been approved as it satisfies the academic requirements in respect of the project work prescribed for the said degree.

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UNDERTAKING

We, **Anitha N. K. (VAS23CE011), Anupa V. J. (LVAS23CE048), K Ashika Musthafa (VAS23CE023) and M Sidharth Anand (VAS23CE028)**, hereby undertake that the project Phase-I report titled **APPLICATION OF COIR GEOTEXTILE FOR SLOPE STABILIZATION IN EMBANKMENT** is carried out by us independently under the guidance of **Dr Abhilasha P. S.** (Head of Department, Dept. of CE) and **Ms Nirmala Krishnan** (Asst. Prof., Dept. of CE), Vidya Academy of Science & Technology, Thrissur, in partial fulfilment of the requirements for the award of the **Degree of Bachelor of Technology** in **Civil Engineering** of **APJ Abdul Kalam Technological University**, during the academic year 2026-2027.

Place: Thrissur
Date: October 2026

Anitha N. K. (VAS23CE011)

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Vision and Mission

Vidya Academy of Science & Technology

Vision

To be a premier institution in engineering education and research, nurturing innovative professionals for sustainable societal development.

Mission

M1: Provide quality engineering education that empowers students with strong technical competence and analytical problem-solving abilities.

M2: Promote research, innovation, entrepreneurship and industry collaboration to develop sustainable technological solutions.

M3: Develop ethical and socially responsible professionals with leadership qualities, lifelong learning abilities and a commitment to societal well-being.

Department of Civil Engineering

Vision

To excel in Civil Engineering education and research, moulding innovative and socially responsible professionals for sustainable development.

Mission

M1: Provide quality education in Civil Engineering that imparts technical knowledge and develops problem-solving skills.

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M3: Develop professionally competent Civil Engineers with ethical values, leadership qualities and commitment for societal well-being.

Program Educational Objectives (PEOs)

PEO-1: Graduates will be able to apply knowledge of Civil Engineering principles, analytical skills, and modern engineering tools to provide innovative solutions for complex engineering problems.

PEO-2: Graduates will be able to excel in professional careers and higher studies by demonstrating technical competence, research aptitude, leadership qualities, and effective communication skills.

PEO-3: Graduates will be able to contribute to sustainable infrastructure development through innovation, entrepreneurship, and application of emerging technologies in Civil Engineering.

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Program Specific Outcomes (PSOs)

PSO-1: Analyze and design structural systems, geotechnical foundations, and infrastructure components using modern design codes and engineering software, ensuring structural safety and sustainability.

PSO-2: Plan and execute construction and infrastructure projects using digital construction technologies, surveying and GIS tools, and project management practices for timely, cost-effective, and quality outcomes.

PSO-3: Innovate in emerging areas such as green building technologies, sustainable water resource management, and smart infrastructure systems to address societal needs.

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Engineering Graduates will be able to:

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design systems/components/processes to meet identified needs with consideration for public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data to provide valid conclusions. (WK8)

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, recognising their limitations to solve complex engineering problems. (WK2 and WK8)

PO6: The Engineer and The World: Analyse and evaluate societal and environmental aspects while solving complex engineering problems for their impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5 and WK7)

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national and international laws. (WK9)

PO8: Individual and Collaborative Team Work: Function effectively as an individual, and as a member or leader in diverse/multidisciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, and make effective presentations considering cultural, language and learning differences.

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making to one's own work, as a member and leader in a team, and to manage projects in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. (WK8)

ACKNOWLEDGEMENT

During the course of our project work, several persons collaborated directly and indirectly with us. Without their support it would be impossible for us to do our work. That is why we wish to dedicate this section to recognize their support.

First and foremost, we thank God Almighty for His providence and for being the guiding light throughout the project.

We would like to thank our guide Dr Abhilasha P. S., Head of the Civil Engineering Department, and co-guide Ms Nirmala Krishnan, Assistant Professor, Civil Engineering Department, for providing critical inputs in the preparation of this report. We also thank all other faculty members in our department for their guidance.

We are thankful to Dr Abhilasha P. S., Head of Civil Department, and our Principal Dr Sunitha C., for their cooperation.

Finally, we extend our sincere gratitude to our friends who have always been helpful in preparing and presenting the report and in the discussion following the presentation.

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SUSTAINABLE DEVELOPMENT GOALS (SDGs)

1. **No Poverty:** End poverty in all its forms everywhere.
2. **Zero Hunger:** End hunger, achieve food security and improved nutrition, and promote sustainable agriculture.
3. **Good Health and Well-being:** Ensure healthy lives and promote well-being for all at all ages.
4. **Quality Education:** Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.
5. **Gender Equality:** Achieve gender equality and empower all women and girls.
6. **Clean Water and Sanitation:** Ensure availability and sustainable management of water and sanitation for all.
7. **Affordable and Clean Energy:** Ensure access to affordable, reliable, sustainable and modern energy for all.
8. **Decent Work and Economic Growth:** Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.
9. **Industry, Innovation and Infrastructure:** Build resilient infrastructure, promote inclusive and sustainable industrialisation and foster innovation.
10. **Reduced Inequalities:** Reduce inequality within and among countries.
11. **Sustainable Cities and Communities:** Make cities and human settlements inclusive, safe, resilient and sustainable.
12. **Responsible Consumption and Production:** Ensure sustainable consumption and production patterns.
13. **Climate Action:** Take urgent action to combat climate change and its impacts.
14. **Life Below Water:** Conserve and sustainably use the oceans, seas and marine resources.
15. **Life on Land:** Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, halt and reverse land degradation and halt biodiversity loss.
16. **Peace, Justice and Strong Institutions:** Promote peaceful and inclusive societies, provide access to justice for all and build effective, accountable and inclusive institutions at all levels.
17. **Partnerships for the Goals:** Strengthen the means of implementation and revitalise the global partnership for sustainable development.

ABSTRACT

Embankment and riverbank slope failures may result from the combined effects of soil properties, water movement, surface erosion and inadequate drainage. This project examines the potential use of coir geotextile for slope stabilization in an embankment, taking the bund failure site along the Kurumali River at Rappal, Thrissur, as the field reference. A site visit was conducted, and three soil samples were collected from different heights of the failed slope. Preliminary laboratory investigations included water-content determination, liquid-limit observations and sieve analysis. The recorded water contents for Samples 1, 2 and 3 were 20%, 33.33% and 25%, respectively; the value for Sample 1 was recalculated using its recorded masses. The liquid-limit and particle-size parameters are treated as preliminary observations and require confirmation using the original laboratory records and graphs.

The next phase, planned for Semester VIII, will include plastic-limit, direct-shear and California Bearing Ratio (CBR) tests. Based on the soil characteristics and project requirements, a suitable coir geotextile will be selected using its mass per unit area and relevant performance properties. A Gradient Ratio Test and slope-stability analysis using GeoStudio are also planned. The stability of unreinforced and coir-geotextile-reinforced conditions will be compared after the necessary soil and geotextile parameters have been established. Since these investigations have not yet been completed, this Phase-I report presents the field activities and preliminary test observations only; it does not claim a measured improvement in stability or report a final factor of safety.

Keywords: coir geotextile, embankment stability, slope stabilization, soil testing, GeoStudio

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LIST OF SYMBOLS AND ABBREVIATIONS

Abbreviation	Description
CBR	California Bearing Ratio
C_c	Coefficient of curvature
C_u	Coefficient of uniformity
GRT	Gradient Ratio Test
GSM	Grams per square metre
LL	Liquid limit
D_{10}, D_{30}, D_{60}	Particle sizes at 10%, 30% and 60% finer
w	Water content (%)

Chapter 1

INTRODUCTION

Road embankments elevate roadways above surrounding terrain. Their stability depends on soil strength, compaction, drainage, geometry and the condition of the foundation soil. In wet conditions, increased pore-water pressure can reduce effective stress and shear strength, while runoff and seepage can erode exposed slopes.

This project studies the application of coir geotextile for slope stabilization in embankments. The motivating field location is the bund failure site along the Kurumali River at Rappal, Thrissur, reported on 9 June 2026. The investigation considers the soil conditions at the site and the potential use of coir geotextile as a natural reinforcement and erosion-control material.

1.1 Geotextiles

Geotextiles are permeable textile materials used in contact with soil or other geotechnical materials. Depending on their construction and application, they may provide separation, filtration, drainage, reinforcement or erosion control. Synthetic geotextiles are commonly manufactured from polymers such as polypropylene and polyester. Natural-fibre alternatives include coir and jute.

1.2 Coir Geotextile

Coir is obtained from coconut husk. Coir geotextiles can protect exposed soil, reduce surface erosion and provide tensile reinforcement. Their open structure allows water movement and can support vegetation establishment. Performance depends on fibre quality, weave, mass per unit area (GSM), installation, soil properties and environmental exposure.

1.3 Objectives

1. To investigate the engineering properties of soil collected from the selected site.
2. To evaluate the slope stability of the embankment.
3. To study the application of coir geotextile in the soil embankment.
4. To develop a predictive model in GeoStudio to simulate performance and service-life behaviour.



Figure 1: Coir geotextile

1.4 Scope of the Study

The scope includes characterization of site soil through laboratory tests, evaluation of strength and shear characteristics, permeability-related assessment of the soil–geotextile system, comparison of reinforced and unreinforced conditions, and GeoStudio slope-stability analysis. Environmental factors affecting geotextile strength, durability and service life are also considered.

1.5 Sustainable Development Goals

The project is associated with SDG 9 (Industry, Innovation and Infrastructure), SDG 11 (Sustainable Cities and Communities), SDG 12 (Responsible Consumption and Production) and SDG 13 (Climate Action), through its focus on resilient infrastructure, locally available natural materials and rainfall-related slope performance.

1.6 Materials and Equipment

The materials are soil samples collected from the selected site and coir geotextile. The proposed equipment includes a direct shear apparatus, sieve set, Casagrande liquid-limit apparatus and Gradient Ratio Test (GRT) apparatus. GeoStudio software is proposed for slope-stability modelling.

1.7 Soil Tests

Sieve analysis is used to determine particle-size distribution and obtain D10, D30 and D60 where the curve supports these readings. The coefficients are Cu=D60/D10 and Cc=$D30^2$/(D60D10). The liquid limit is determined using the Casagrande apparatus as the water content corresponding to 25 blows. Plastic limit is the water content at which rolled soil threads begin to crumble. Water content is calculated from the mass of water divided by the mass of dry soil.

1.8 Future Scope

Future work includes long-term durability studies, additional embankment geometries and groundwater conditions, comparison with field monitoring or physical model tests, and assessment of the effects of fibre quality, thickness, tensile properties and biodegradation.

Chapter 2

LITERATURE REVIEW

The literature reviewed in this project addresses natural geotextiles, soil reinforcement, road embankments, filtration, erosion control and drainage. The studies are summarized in Table 2.1.

Table 2.1: Summary of literature review

Sl. No.	Author and Year	Study / Topic	Objective / Focus	Methodology / Aspect	Key Findings
1	K. S. Beena (2013)	Case studies on coir and geotextiles for soil stabilization	Use of coir geotextiles for soil stabilization	Case studies	Coir is potentially effective, sustainable and economical.
2	Ghosh et al. (2019)	Biodegradability of jute geotextiles for road applications	Study of the service life of jute geotextiles	Influence of soil properties and chemical treatment	Soil properties and chemical treatment influence biodegradability and service life.
3	Meshram et al. (2013)	Application of coir geotextile in rural roads on black-cotton soil subgrade	Use of coir geotextile in rural road construction	Application on black-cotton soil subgrade	Reports improved subgrade performance.
4	Ogundare et al. (2018)	Utilization of geotextile for soil stabilization	Evaluate geotextile for soil and pavement improvement	Soil stabilization applications	Geotextiles can improve subgrade strength and pavement performance.
5	Arsyad et al. (2020)	Full-scale road embankment investigation on soft soil with geotextile stabilization	Assess geotextile-stabilized embankments on soft soil	Field investigation and numerical analysis	Examines field and numerical performance of stabilized embankments.
6	Rajagopal (2009)	Coir geotextile as a separation and filtration layer for low-intensity road bases	Study separation and filtration applications	Field applications in low-intensity road bases	Considers cost-effective applications of coir geotextile.

Sl. No.	Author and Year	Study / Topic	Objective / Focus	Methodology / Aspect	Key Findings
7	Sayida and Evangeline (2020)	Durability of coir geotextile embedded in different subgrade soils	Evaluate durability in different soil conditions	Study of soil type and biodegradation	Soil type and biodegradation influence durability.
8	Vishwanathan (2020)	Field evaluation of coir-geotextile-reinforced subgrade for low-volume pavements	Evaluate reinforced subgrade performance	Field evaluation and physical properties	Discusses the role of physical properties in subgrade performance.
9	Basu et al. (2009)	Construction of unpaved rural roads using jute–synthetic blended woven geotextile	Study blended geotextiles in rural roads	Use of jute–synthetic woven geotextile	Reports improved mechanical performance of the blend.
10	Vishnudas and Savenji (2006)	Vegetated embankment covering using coir geotextile	Study erosion protection for embankments	Combination of coir geotextile and local grass species	Discusses vegetation and coir geotextile for erosion protection.
11	Anushitha et al. (2020)	Performance of coir-geotextile-reinforced subgrade for low-volume roads	Evaluate reinforced subgrade performance	Study of coir-geotextile reinforcement	Considers improvement in bearing capacity.
12	Mukherjee (2014)	Highway surface drainage and waterlogging	Examine highway drainage measures	Surface slopes, inlets, channels and culverts	Outlines measures to reduce waterlogging.
13	Patil and Sharma (2017)	Flexible pavement design in waterlogged areas	Consider pavement performance in waterlogged conditions	Subgrade stabilization and drainage	Highlights the importance of subgrade stabilization and drainage.
14	Radhakrishnan (2025)	SANDRP article on NH66 in Kerala	Discuss embankment construction, drainage and flooding	Contextual discussion of NH66 in Kerala	Provides background on the relationship between embankment construction, drainage and flooding. This is not the study location of the present project.

The reviewed work supports examining coir geotextile for reinforcement, filtration and erosion control. However, its performance at the selected Rappal site must be established using site-specific laboratory results and stability analysis.

Chapter 3

METHODOLOGY

The methodology consists of field investigation, collection of soil samples, preliminary laboratory testing, further strength and permeability testing, selection of coir geotextile and numerical slope-stability analysis.

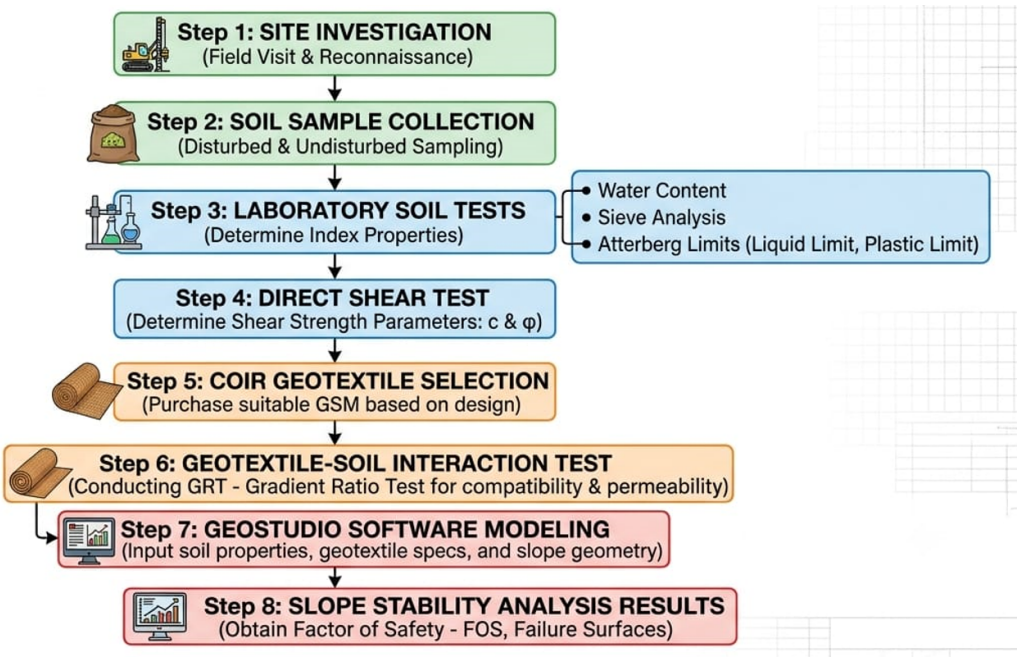


Figure 2: Flow chart of methodology

3.1 Site Investigation

The bund failure site along the Kurumali River at Rappal, Thrissur, was visited. The event and site conditions were discussed with the Assistant Engineer associated with the location.

3.2 Soil Sample Collection

Three soil samples were collected from different heights of the failed slope to examine possible variation in soil properties with location.

3.3 Preliminary Laboratory Tests

Water-content determination, liquid-limit testing and sieve analysis were included in the preliminary investigation. Plastic-limit testing is also listed in the project programme.

3.4 Direct Shear Test

The direct shear test is planned to determine shear-strength characteristics. Normal stress and measured shear response will be used to estimate cohesion and angle of internal friction for subsequent analysis.

3.5 Coir Geotextile Selection

A suitable coir geotextile GSM is to be selected based on the proposed test programme, including CBR results and the required application performance.

3.6 Gradient Ratio Test

The GRT is planned to evaluate soil–geotextile interaction with respect to flow and potential clogging behaviour. The procedure and test conditions should be documented, and observations compared for the selected soil and geotextile.

3.7 GeoStudio Modelling

A numerical model is to be developed in GeoStudio using the soil parameters obtained from laboratory testing. Embankment geometry, groundwater or pore-water conditions and geotextile representation must be documented.

3.8 Slope-Stability Analysis

The analysis will evaluate the embankment under defined conditions and compare reinforced and unreinforced cases. Numerical results are pending completion of the planned tests and modelling.

Chapter 4

PROJECT STATUS COMPLETED IN S7

4.1 Site Investigation

The project team visited the bund failure location along the Kurumali River at Rappal, Thrissur. The site conditions and the reported failure were discussed with the Assistant Engineer associated with the location. The visit helped the team observe the failed embankment and understand the field context.



Figure 3: Site investigation; Kurumali River Bund, Rappal, Thrissur, Kerala

4.2 Soil Sample Collection

Three soil samples were collected from different heights of the failed slope to examine possible variations in soil properties with location.



Figure 4: Collection of soil sample from Kurumali River Bank

4.3 Preliminary Laboratory Tests

The preliminary laboratory work completed in Semester VII included water-content determination, liquid-limit observations and sieve analysis. The recorded observations and calculations are presented in Chapter 5. Sample 1 water content is recalculated as 20% from its recorded masses; the original masses are retained. The liquid-limit and particle-size parameters remain preliminary and should be checked against the original graphs and complete laboratory records.

Chapter 5

RESULTS AND DISCUSSION

5.1 Water Content

Water content is calculated as

$$w = \frac{W_2 - W_3}{W_3 - W_1} \times 100$$

where W_1 is the mass of the empty container, W_2 is the mass of container plus wet soil, and W_3 is the mass of container plus dry soil.

Table 5.1: Water-content observations

Sample	W_1 (g)	W_2 (g)	W_3 (g)	Water content (%)
1	20	50	45	20.00
2	20	40	35	33.33
3	20	45	40	25.00

For Sample 1, water mass is $50-45=5$ g and dry-soil mass is $45-20=25$ g; therefore water content is $5/25 \times 100 = 20\%$. This corrects the 40% value printed in the source table, while retaining its recorded masses.

5.2 Liquid Limit

Table 5.2: Liquid-limit observations

Determination	1	2	3
Container mass (g)	20	20	20
Container + wet soil (g)	49	50	51
Container + dry soil (g)	40	40	40
Water content (%)	45	50	55
Number of blows	30	28	27

The source report gives a liquid limit of 56% at 25 blows and a flow index of 218.54. These are retained as reported values, but the flow curve and calculation should be verified from the original graph and test record.

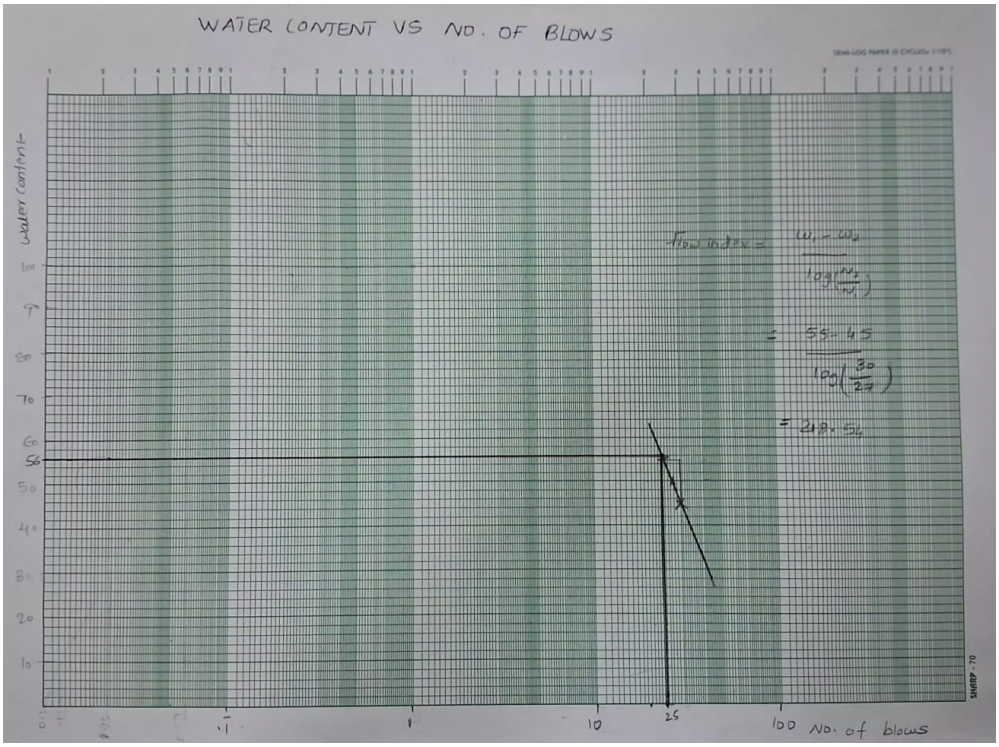


Figure 5: Liquid-limit graph

5.3 Sieve Analysis

Table 5.3: Sieve-analysis observations

Sieve opening (mm)	Mass retained (g)	Retained (%)	Cumulative (%)	Finer (%)
1.18	700	70	70	30
0.600	240	24	94	6
0.425	40	4	98	2
0.300	10	1	99	1
0.075	10	1	100	0

The source report states $D_{10} = 0.68$, $C_u = 3.82$ and $C_c = 0.787$, and describes the soil as sandy. These reported values are reproduced as source values, not independently validated: the listed sieve table does not show sufficient data to verify the complete particle-size curve or D_{60}. The soil classification and coefficients should therefore be checked against the original graph and complete sieve record.

5.4 Discussion

The three samples show different moisture contents, with Sample 1 equal to 20% after recalculation. The liquid-limit and sieve-analysis results are preliminary. Direct shear, CBR, GRT, geotextile selection and GeoStudio modelling remain to be completed in Semester VIII; consequently, no final reinforced-versus-unreinforced stability result is claimed.

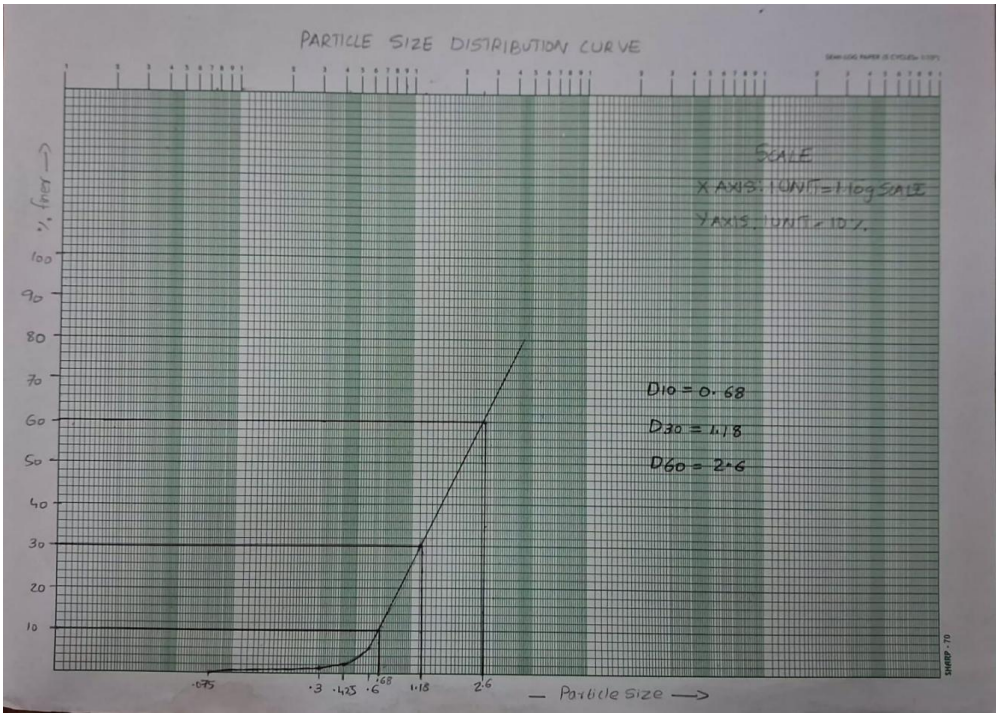


Figure 6: Particle-size distribution curve

Chapter 6

FUTURE WORK AND ACTIVITIES PLANNED FOR SEMESTER VIII

The following activities are planned for completion in Semester VIII:

1. Conduct the direct shear test to determine the shear-strength parameters of the collected soil.
2. Conduct the California Bearing Ratio (CBR) test.
3. Select a suitable coir geotextile based on its GSM, relevant properties, soil characteristics and project requirements.
4. Conduct the Gradient Ratio Test (GRT) to examine the soil–geotextile interaction and permeability-related behaviour.
5. Carry out slope-stability analysis using GeoStudio with the required soil and geotextile parameters.
6. Study and evaluate the results, including comparison of reinforced and unreinforced embankment conditions.

These activities are pending. No final test results or factor of safety are reported in this Phase-I document.

Chapter 7

CONCLUSION

The present project, entitled "Application of Coir Geotextile for Soil Embankment Stabilization," focuses on investigating the potential of natural coir geotextiles in improving the stability of soil embankments and controlling soil erosion. Embankment failures are major geotechnical concerns, particularly in regions like Kerala, where heavy rainfall, waterlogging, poor drainage conditions and variations in soil properties can affect the strength and stability of soil slopes. The study explores a sustainable and locally available reinforcement material for suitable geotechnical applications.

As part of the project, a site investigation was conducted at the bund failure location along the Kurumali River at Rappal, Thrissur. The site visit provided an opportunity to observe the field conditions and understand the reported embankment failure. Discussions with the Assistant Engineer associated with the location provided additional information about the site. Three soil samples were collected from different heights of the failed slope to investigate the engineering properties of the soil and examine possible variations in soil characteristics at different locations.

Preliminary laboratory tests were conducted on the collected soil samples, including water-content determination, liquid-limit observations and sieve analysis. The water-content test indicated moisture contents of 20%, 33.33% and 25% for the three samples, respectively. These variations indicate differences in the moisture conditions of the collected samples. The liquid limit was reported as 56%. The source report describes the soil as predominantly sandy and reports a coefficient of uniformity (C_u) of 3.82 and a coefficient of curvature (C_c) of 0.787. These are preliminary reported values and should be verified against the original laboratory records and graphs before being treated as confirmed results.

The literature review highlights the potential applications of coir geotextiles in soil reinforcement, erosion control, filtration and drainage. Coir is a natural fibre obtained from coconut husk and is widely available in Kerala. Its porous structure, tensile characteristics and biodegradable nature make it a promising material for suitable geotechnical applications. Previous studies indicate that coir geotextiles may help improve soil performance, protect exposed embankment surfaces against erosion and support sustainable construction practices. Their effectiveness and service life depend on factors such as soil type, moisture conditions, fibre quality, installation methods and environmental exposure.

The proposed application of coir geotextile at the selected embankment is intended to provide surface protection, reduce soil erosion and improve soil–geotextile interaction. Selecting a suitable geotextile based on its mass per unit area (GSM), strength and other relevant physical properties is important for its performance. Further investigations, particularly direct shear testing and California Bearing Ratio (CBR) testing, are required to obtain the necessary soil-strength characteristics and support the selection of a suitable coir geotextile.

The project also proposes the use of GeoStudio software for slope-stability analysis. Using the soil parameters obtained from laboratory investigations, the stability of the embankment will be evaluated under different conditions. A comparison between reinforced and unreinforced

soil conditions will help assess the contribution of coir geotextile to embankment stability. The planned Gradient Ratio Test (GRT) will also provide information about soil–geotextile interaction and permeability-related behaviour.

In conclusion, the Phase-I work has established the field background, completed soil sample collection and carried out preliminary laboratory investigations. Further laboratory tests, selection of a suitable coir geotextile and numerical slope-stability analysis remain to be completed in Semester VIII. The findings from these activities will help evaluate the suitability and effectiveness of coir geotextile for the selected embankment and its potential contribution to erosion control and soil stabilization using sustainable, locally available materials.

7.1 Mapping with POs and PSOs

The mapping below reflects the work completed in Phase I. Levels are indicative and should be confirmed by the project guide and department before submission. Mapping level: 1 – Low, 2 – Medium, 3 – High.

Table 7.1: Mapping of the project with POs and PSOs

PO / PSO	**Level**	**Justification**
PO1: Engineering Knowledge	2	Applied soil mechanics and geotechnical engineering concepts to the project.
PO2: Problem Analysis	2	Examined the failed embankment context and documented preliminary soil-test observations.
PO3: Design/Development of Solutions	1	Coir-geotextile application and reinforced design comparison are planned for the next phase.
PO4: Conduct Investigations of Complex Problems	2	Conducted a site visit, collected soil samples and carried out preliminary laboratory tests.
PO5: Engineering Tool Usage	1	GeoStudio modelling is planned; it has not yet been completed.
PO6: The Engineer and the World	2	The study considers resilient embankments, erosion control and the use of a natural-fibre material.
PO7: Ethics	1	Test observations and limitations are reported without presenting pending results as completed.
PO8: Individual and Collaborative Team Work	2	The project work is undertaken collaboratively by a four-member student team.
PO9: Communication	2	The site observations, methodology and preliminary results are documented in this report.
PO10: Project Management and Finance	1	The project is organised in semester-wise phases with further testing and modelling planned.

PO / PSO	Level	Justification
PO11: Life-Long Learning	2	The work involves learning and applying geotechnical laboratory procedures and numerical analysis methods.
PSO-1: Structural systems, geotechnical foundations and infrastructure design	2	The project investigates soil properties and embankment slope stabilization.
PSO-2: Construction and infrastructure project planning and execution	1	Site investigation and laboratory planning support infrastructure assessment; project execution tools remain future work.
PSO-3: Emerging and sustainable Civil Engineering technologies	2	The project examines coir geotextile as a natural material for sustainable geotechnical applications.

7.2 Mapping with SDGs

The project is relevant to the following Sustainable Development Goals through its focus on resilient infrastructure, slope protection and a renewable natural-fibre material.

Table 7.2: Mapping of the project with SDGs

SDG	Contribution of the project
SDG 9: Industry, Innovation and Infrastructure	Investigates a geotechnical approach that may contribute to more resilient embankment infrastructure.
SDG 11: Sustainable Cities and Communities	Addresses slope and embankment stability, which are relevant to safer and more resilient communities.
SDG 12: Responsible Consumption and Production	Studies coir, a renewable coconut-husk fibre, as a potential geotechnical material.
SDG 13: Climate Action	Considers erosion and water-related slope performance under environmental and rainfall conditions.

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APPENDIX

Appendix A: Site Investigation

A site visit was conducted at the failed bund along the Kurumali River at Rappal, Thrissur, Kerala. The site conditions and reported failure were observed, and the field context was discussed with the Assistant Engineer.

Three soil samples were collected from different heights of the failed slope:

- Sample 1: Upper portion of the slope
- Sample 2: Middle portion of the slope
- Sample 3: Lower portion of the slope

Appendix B: Preliminary Laboratory Tests

The preliminary laboratory tests completed in Semester VII included water-content determination, liquid-limit observations and sieve analysis. The observations and calculations are presented in Chapter 5.

Appendix C: Summary of Preliminary Results

- Water content: Sample 1 – 20%, Sample 2 – 33.33%, Sample 3 – 25%.
- Reported liquid limit: 56%.
- Reported flow index: 218.54.
- Reported particle-size parameters: $D_{10} = 0.68$ mm, $C_u = 3.82$ and $C_c = 0.787$.
- The soil was described as sandy in the source report.

The liquid-limit and particle-size parameters are retained as reported preliminary values and require verification against the original laboratory graphs and records.

Appendix D: Photographic Documentation

Photographs of the site investigation and soil sample collection are included in Chapter 4.

Appendix E: Further Laboratory Investigations

The following tests are planned for Semester VIII:

- Direct shear test
- California Bearing Ratio (CBR) test
- Gradient Ratio Test (GRT)
- Plastic-limit determination

Appendix F: Coir Geotextile Selection

A suitable coir geotextile will be selected based on the soil characteristics and project requirements. Its relevant properties, including mass per unit area (GSM), will be considered during the next phase.

Appendix G: Slope Stability Analysis

GeoStudio modelling is planned for Semester VIII to assess the embankment stability. The stability of unreinforced and coir-geotextile-reinforced conditions will be compared after the required soil and geotextile parameters are obtained.

Appendix H: Project Work Status

- Semester VII: Site investigation, soil sample collection and preliminary laboratory tests completed.
- Semester VIII: Further laboratory tests, coir geotextile selection, GRT and GeoStudio analysis planned.

Appendix I: Supporting Documents

Original laboratory observation sheets, test graphs and relevant equipment datasheets may be added when available and verified.



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